

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459306.6388 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.179 +/- 0.019
Transit depth (Rp/Rs)^2	3.2 +/- 0.68 [%]
Orbital Inclination (inc)	86.99 +/- 1.73
Airmass coefficient 1 (a1)	0.8718 +/- 0.0082
Airmass coefficient 2 (a2)	0.101 +/- 0.046
Scatter in the residuals of the lightcurve fit is	0.94 %
Best Comparison Star	#2 - [415, 64]
Optimal Aperture	4.44
Optimal Annulus	16.31
Transit Duration (day)	0.0385 +/- 0.0021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.179 $\pm$ 0.019 vs 0.1594 (deviation 0.93 σ)
Duration fit vs published	0.0385 d vs 0.0512 d (deviation 5.49 σ)
Mid-time O-C	-5.0 min
Depth SNR	8.5
Residual scatter	0.94 %

Light curve

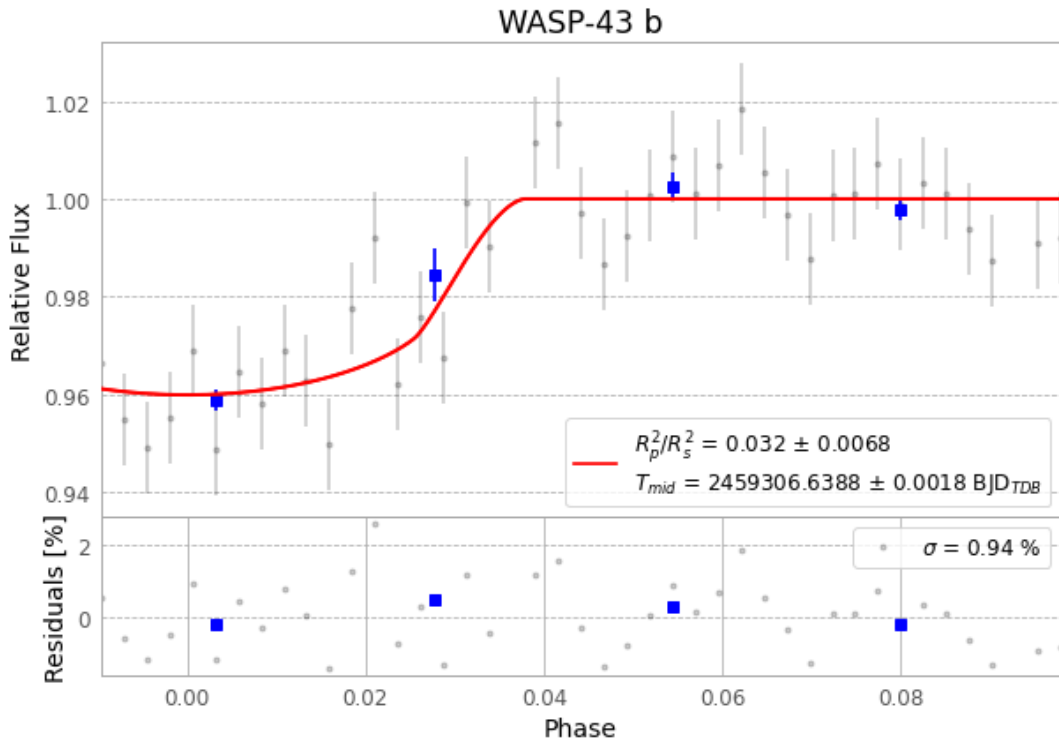


Figure 1 — FinalLightCurve_WASP-43 b_2021-04-01.png (SHA-256 5ccab247fe2e3304...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [482, 67], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 def67a6e7b215a03...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

43 FITS frames (28 MB), WASP-43210402030012.FITS → WASP-43210402050612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-210402.log (2f6208e2f918...), FinalLightCurve_WASP-43 b_2021-04-01.png (5ccab247fe2e...), FinalLightCurve_WASP-43 b_2021-04-01.csv (1abd59a43923...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 11:31Z.